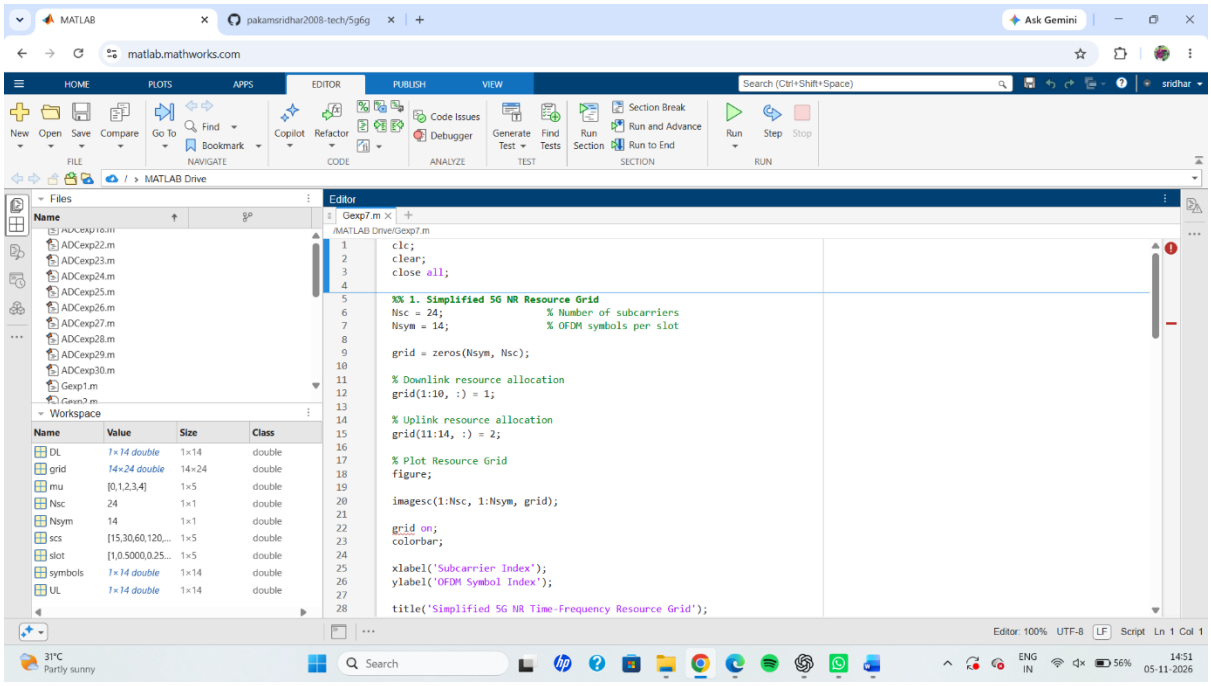
Experiment : 7

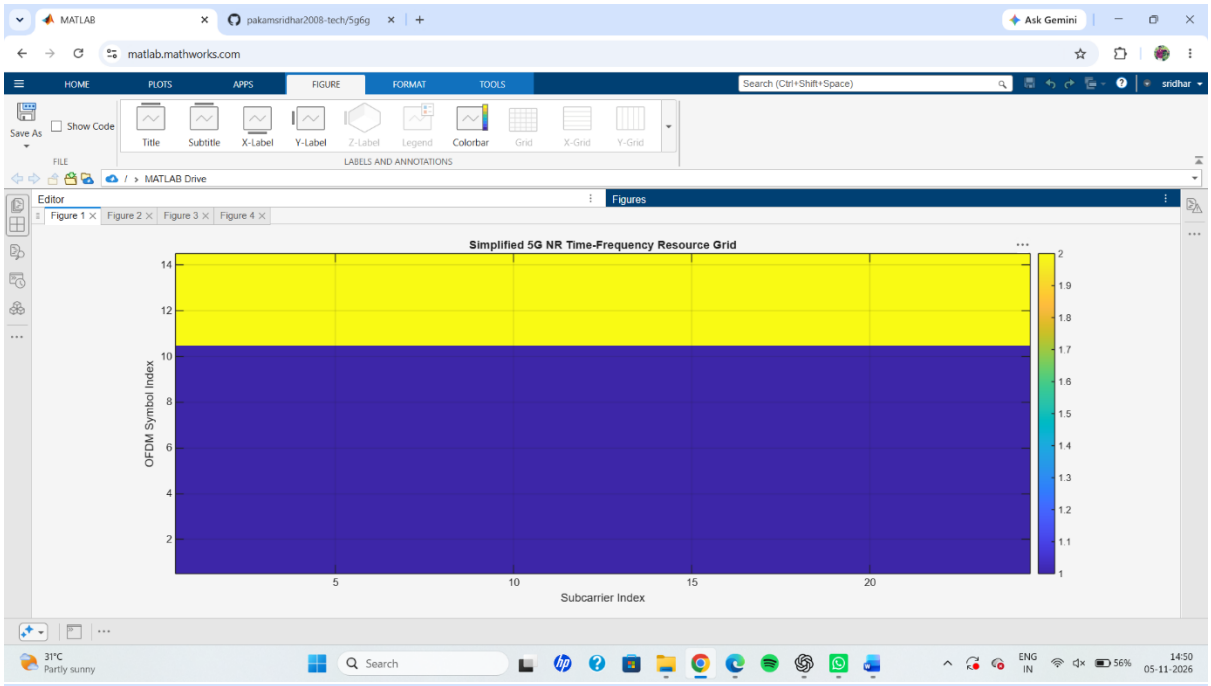
# Analysis of 5G NR Frame Structure and Subcarrier Spacing

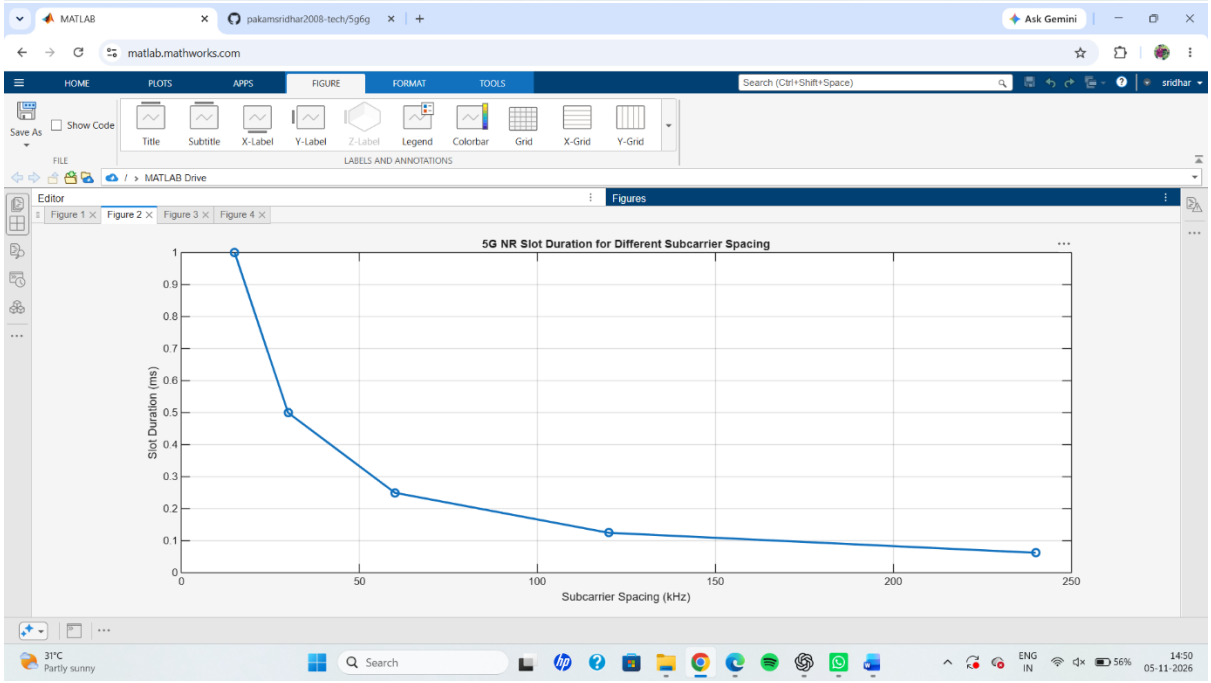
OBJECTIVES :

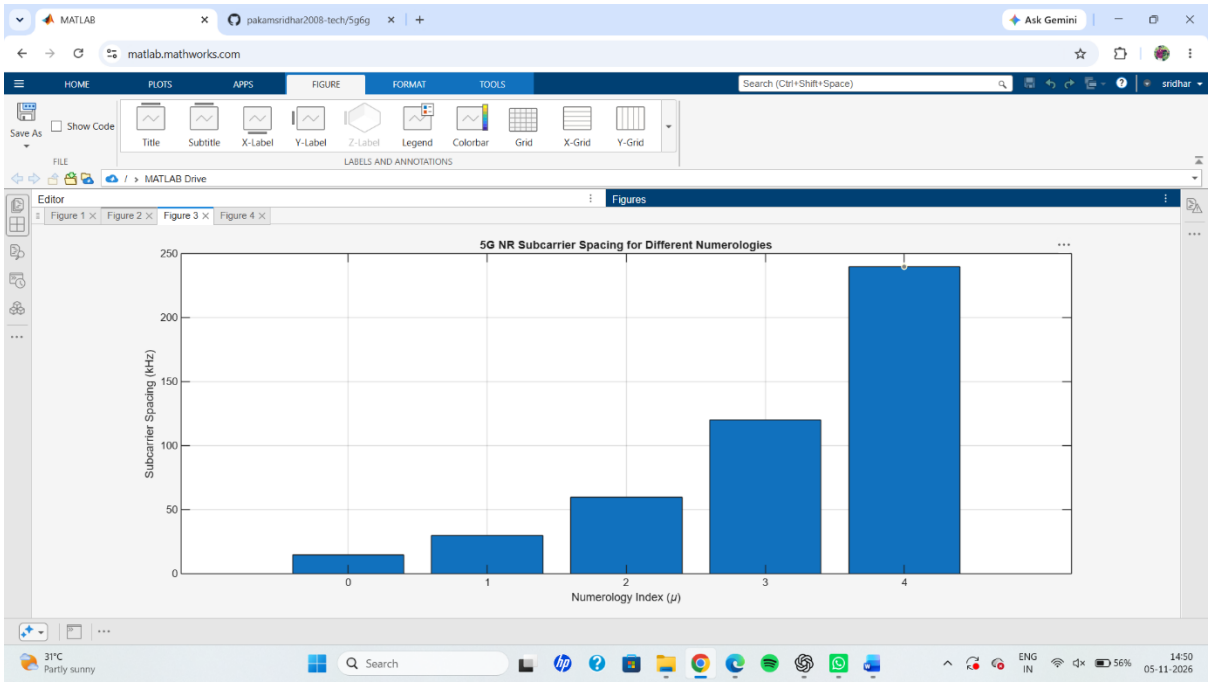
- To generate or inspect a simplified 5G NR resource grid.
- To compare selected subcarrier spacings and their effect on symbol or slot duration.
- To identify basic downlink and uplink resource assignments in the model.
- To relate the radio resources to gNodeB functions, RAN organization and RRC's role.

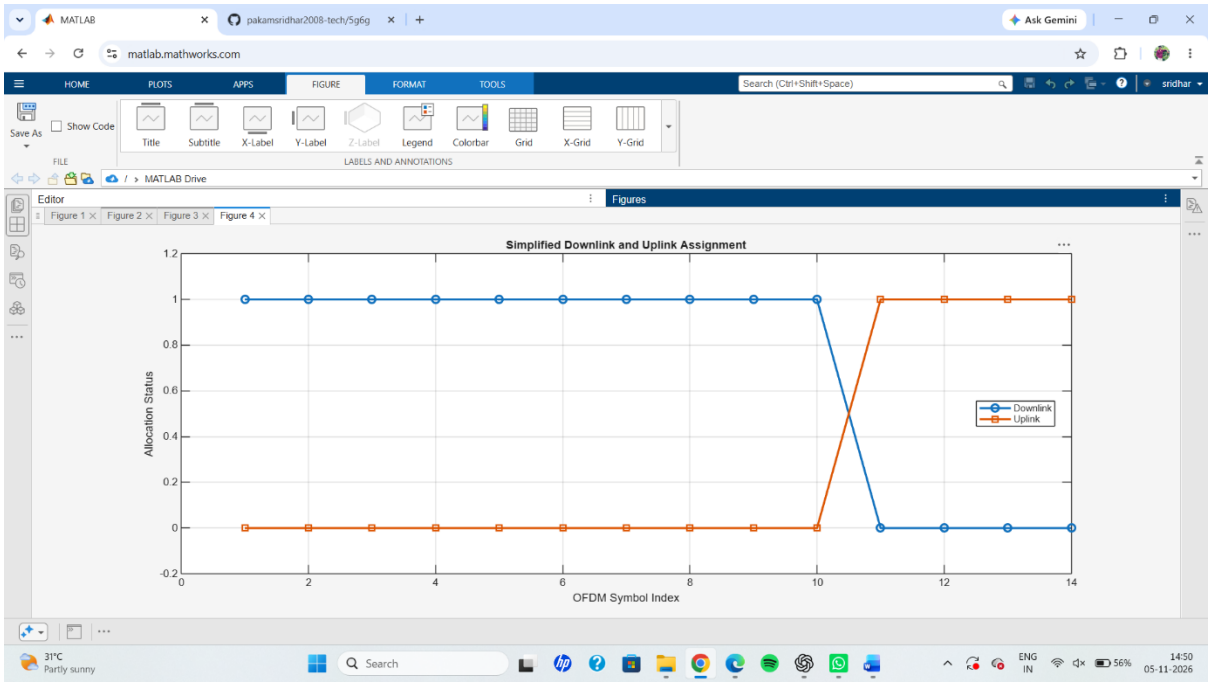
PROGRAME CODE :




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pakamsridhar2008-tech/5g6g
Ask Gemini
matlab.mathworks.com
HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
Search (Ctrl+Shift+Space)
sridhar
Save As
Show Code
FILE
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
LABELS AND ANNOTATIONS
/ › MATLAB Drive
Editor
Figures
Figure 1
Figure 2
Figure 3
Figure 4
Simplified 5G NR Time-Frequency Resource Grid
OFDM Symbol Index
Subcarrier Index
31°C
Partly sunny
Search
ENG
IN
56%
14:50
05-11-2026


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X-Grid
Y-Grid
LABELS AND ANNOTATIONS
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Figures
Figure 1
Figure 2
Figure 3
Figure 4
5G NR Slot Duration for Different Subcarrier Spacing
Slot Duration (ms)
Subcarrier Spacing (kHz)
31°C
Partly sunny
Search
ENG
IN
56%
14:50
05-11-2026




RESULT :

1. A simplified 5G NR resource grid was generated and inspected.

2. The effect of different SCS values on slot duration was analyzed.

3. Basic downlink and uplink resource assignments were represented.

4. The relationship among NR radio resources, gNodeB operation, RAN organization and RRC was explained.